

Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; RPN, risk priority number; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This document records the risk assessment made on the A-Mab drug substance process before any characterization study was executed. Its output is the scope of the characterization campaign. For every process parameter in the transferred process description it names the quality attributes the parameter could affect, the potential failure mode and its effect, an initial risk score, and the type of study assigned to the parameter. The assessment is prospective throughout and reports no result.

The assessment does not classify parameters. A parameter is classified as a CPP, a WC-CPP, a key process parameter or a GPP from the effects the characterization studies measure, under SOP-4001, and the classification is reported in the per unit operation reports and consolidated in PCMR-001. The scores given below are initial scores in the same sense. The risk that remains once the studies have reported is a separate assessment made against data that does not yet exist, and nothing in this document anticipates it.

The scope is the drug substance process from the production bioreactor to formulation. Table 1.1 lists the 8 unit operations and the role each has in the control strategy. In total 37 process parameters are carried into the assessment across those unit operations. The seed expansion train, drug product manufacture and the analytical methods themselves lie outside the scope of this document.

Table 1.1: The A-Mab drug substance process train and the role of each unit operation in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance

Step	Unit operation	Principal role
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

PTP-001 is the parent of this assessment and defines the transfer it supports. Table 1.2 lists the documents this assessment sits between. Its consumers are the characterization plans, one for each unit operation, which take the study type assigned here and set out how that study will be run.

Table 1.2: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Risk assessment in this framework is a lifecycle activity and not a single exercise (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). This assessment is the first in that lifecycle and draws on prior knowledge alone, since no A-Mab characterization data exist at the time it is made. It is revisited when the characterization reports are available, and again once commercial scale experience has been gained. The approach follows the enhanced development approach of ICH Q8 and ICH Q11 and supports Stage 1 of the process validation lifecycle (U.S. Food and Drug Administration 2011).

2 Risk assessment methodology

The method is risk ranking and filtering, applied to process parameters as in the A-Mab case study (CMC Biotech Working Group 2009). Each parameter is put to two questions. The first asks what the parameter could do to a quality attribute if it moved outside the range over which the process is currently understood. The second asks whether the parameter could act together with another parameter of the same unit operation, so that the size of its effect depends on where that other parameter sits. The first question sets the score. The second sets the study type, because an effect that depends on a second parameter cannot be found by moving one parameter at a time.

Severity is taken from the attribute at risk and not from the parameter. It runs from 2 to 10 and is the severity assigned to the attribute in the quality attribute assessment, reproduced in Table 3.1. Where a parameter could affect more than one attribute, the severity of the most severe of them is used. A parameter that puts no quality attribute at risk takes a severity of 4, the value assigned to a process attribute such as step yield or titer.

Occurrence and detection are assigned before any effect has been measured, and they are assigned by class rather than parameter by parameter. Occurrence is scored at 7 for a parameter that could affect a quality attribute, because neither the size of its effect nor the width of the range that holds the attribute in limit is known at this point. It is scored at 4 for a parameter that puts no quality attribute at risk, where the platform history bounds how often the parameter leaves its range. Detection is scored on what the existing controls would reveal. A failure that acts on viral clearance is scored at 10, since no in-process control and no release test on the drug substance measures the log reduction a step achieves. A failure that changes another quality attribute is scored at 8, since release testing measures the attribute but monitoring during the unit operation does not. A failure that puts no quality attribute at risk is scored at 6, since it shows up in the performance of a downstream unit operation before the batch is released.

The initial risk priority number is the product of the three scores. Across the register it runs from 96 to 700 and takes 5 distinct values, because occurrence and detection are assigned by class. Severity is therefore what separates one parameter from another within a unit operation. Priority is High where an attribute of severity 8 or above is at risk, Medium where a quality attribute of lower severity is at risk, and Low where no quality attribute is at risk. No parameter is assigned a study type on the number alone.

The assessment assigns 3 study types. A parameter carried into the designed experiment of its unit operation is studied multivariately, which is the only way to estimate an effect that depends on a second parameter. A parameter that could affect a quality

attribute but is not carried into the design is studied one at a time, and the reason it was left out of the design is recorded with it. The A-Mab risk ranking and filtering table permits that substitution where the study can be justified (CMC Biotech Working Group 2009), and the register marks such a parameter univariate (justified). A parameter that puts no quality attribute at risk is studied one at a time. The assignment is the decision this document makes, and each characterization plan states how the assigned study will be executed.

Criticality is treated as a continuum in this framework and not as a binary split (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). The ranking in Table 4.1 therefore orders parameters against each other. It does not divide them into a critical and a non-critical group, and it is not a control strategy. Both of those follow from the studies this assessment scopes.

3 Quality attributes at risk

The severity axis of the assessment comes from the quality attribute register. Table 3.1 gives the 10 drug substance attributes the process must control, their acceptance criteria, the criticality assigned to each and the unit operation that forms it or, for the two cumulative clearance requirements, the step the requirement is assigned to. Criticality is the product of impact and uncertainty, which is Tool #1 of the A-Mab attribute assessment (CMC Biotech Working Group 2009). Attributes of high and very high criticality are called critical, and 6 of the 10 attributes are critical on that definition.

Table 3.1: Drug substance quality attributes, their criticality and the severity carried into this assessment.

Quality attribute	Cate- gory	Acceptance	Criti- cality	Tool #1	Sever- ity	Set by
Afucosylation	glycosy- lation	2-13 % of glycans	H	60	8	Production Bioreactor
Galactosylation (total %Gal)	glycosy- lation	10-40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
High Mannose	glycosy- lation	3-10 % of glycans	VH	80	10	Production Bioreactor
Aggregates (HMW)	size	0-5 % HMW (SEC)	H	60	8	Production Bioreactor
Acidic charge variants (deamidation)	variant charge variant	18-40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36	6	Production Bioreactor
Residual DNA	process impurity	0-0.001 ng/dose	VL	6	2	Production Bioreactor
Leached Protein A	process impurity	0-5 ppm	L-M	16	4	Protein A Chromatogra- phy
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20	10	Anion Exchange Chromatogra- phy

Of the 10 attributes, 7 are formed in the production bioreactor. The remaining 3 are set downstream, one at the capture step and one at each of the steps that fix a cumulative viral clearance requirement. The bioreactor therefore carries more parameters into multivariate study than any other unit operation.

The top severity of 10 is carried by 3 attributes. High mannose carries the highest Tool #1 score in the register at 80. High mannose glycans carry no core fucose, so they raise antibody dependent cell mediated cytotoxicity in the same way afucosylation does, and A-Mab acts on its target through that mechanism. The impact of the attribute is scored on efficacy for that reason, and its uncertainty is scored above that of afucosylation because the relationship rests on external literature rather than on assays run with A-Mab.

Both viral clearance attributes carry the top severity for a different reason. Their requirement is cumulative across the low pH hold, the anion exchange step and the virus filter, and a shortfall at any one of them changes no property of the drug substance that a release test measures (International Council for Harmonisation 2023a). Severity for these two attributes is therefore assigned on the consequence of a shortfall and not on the probability of detecting one.

Acidic charge variants sit at the other end of the register with a Tool #1 score of 4 and a severity of 4. The impact is scored at the bottom of the scale because deamidation of A-Mab occurs at sites outside the binding region and outside the part of the Fc that carries effector function, so the deamidated forms retain their activity. Four bioreactor parameters are nevertheless scored against the attribute, and the range of each over which the acidic variant content stays within its acceptance criterion is one of the things the bioreactor study must establish. Residual DNA is treated the same way. A low severity lowers the priority of a parameter and does not remove it from study.

4 Parameter risk ranking by unit operation

Table 4.1 gives the ranking for the whole train. Of the 37 parameters, 21 are scored against at least one quality attribute and 16 against process performance alone. The ranking puts 16 parameters at High priority, 5 at Medium and 16 at Low. The subsections that follow take each unit operation in turn and give the potential failure mode and effect behind each score, together with the interaction judgement that decides the study type.

Table 4.1: Pre-characterization parameter risk ranking, by unit operation. Scores are initial scores and no residual risk is reported.

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Prior- ity
3	Production Bioreactor	Culture pH	10	560	multivariate DoE	High
3	Production Bioreactor	Culture temperature	8	448	multivariate DoE	High
3	Production Bioreactor	Dissolved CO2 (pCO2)	8	448	multivariate DoE	High
3	Production Bioreactor	Osmolality	8	448	multivariate DoE	High
3	Production Bioreactor	Culture duration	8	448	multivariate DoE	High
3	Production Bioreactor	Dissolved oxygen	4	96	univariate	Low
3	Production Bioreactor	Initial viable cell conc.	4	96	univariate	Low
3	Production Bioreactor	Basal medium concentration	4	96	univariate	Low
3	Production Bioreactor	Nutrient feed-1 volume	4	96	univariate	Low
4	Harvest and Clarification	Centrifugation (rcf)	4	96	univariate	Low
4	Harvest and Clarification	Depth filter load	4	96	univariate	Low
4	Harvest and Clarification	Post-clarification turbidity	4	96	univariate	Low
5	Protein A Chromatography	Protein load	6	336	multivariate DoE	Medium
5	Protein A Chromatography	Elution buffer pH	6	336	multivariate DoE	Medium

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Prior- ity
5	Protein A Chromatography	Load flow rate	4	96	multivariate DoE	Low
5	Protein A Chromatography	End of pool collect	4	96	multivariate DoE	Low
5	Protein A Chromatography	Operating temperature	4	96	univariate	Low
5	Protein A Chromatography	Bed height	4	96	univariate	Low
6	Low-pH Viral Inactivation	Inactivation pH	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	Hold time	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	Temperature	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	A-Mab concentration	4	96	univariate	Low
7	Cation Exchange Chromatography	Protein load	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Load/Wash conductivity	6	336	multivariate DoE	Medium
7	Cation Exchange Chromatography	Elution buffer pH	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Elution stop collect	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Elution flow rate	4	96	univariate	Low
8	Anion Exchange Chromatography	Load pH	10	700	multivariate DoE	High
8	Anion Exchange Chromatography	Equil/Wash-1 conductivity	6	336	multivariate DoE	Medium
8	Anion Exchange Chromatography	Load conductivity	10	700	multivariate DoE	High
8	Anion Exchange Chromatography	Protein load	6	336	multivariate DoE	Medium
8	Anion Exchange Chromatography	Operating flow rate	10	700	univariate (justified)	High
9	Small-Virus Retentive Filtration	Filtration volume (load)	10	700	multivariate DoE	High
9	Small-Virus Retentive Filtration	Filtration pressure	10	700	multivariate DoE	High
10	Ultrafiltration / Diafiltration	Number of diavolumes	4	96	univariate	Low

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Prior- ity
10	Ultrafiltration / Diafiltration	Transmembrane pressure	4	96	univariate	Low
10	Ultrafiltration / Diafiltration	Final DS concentration	4	96	univariate	Low

4.1 Production Bioreactor

The production bioreactor forms the glycan, charge variant and aggregate attributes of the drug substance, so it carries the highest severity of any step in the train. Of its 9 parameters, 5 are scored against a quality attribute, and the highest initial risk priority number in the step is 560.

Culture pH and culture duration are each scored against 5 attributes, more than any other parameter in the register. Culture pH takes a severity of 10 because high mannose is one of the attributes it puts at risk. A culture pH above its range raises high mannose and lowers galactosylation, because a higher extracellular pH raises the pH inside the Golgi and moves the mannosidases and the galactosyltransferase away from the acidic conditions at which they work fastest. Aggregate in the harvest rises with culture pH as well, and afucosylation and the acidic charge variants both fall as it rises.

Culture duration reaches the same number of attributes by a different route. Extending the culture, or harvesting it after viability has fallen, raises host cell protein and DNA in the harvest, because lysing cells release intracellular protein and DNA into the broth. Aggregate rises with culture duration as well. Antibody secreted early in the run stays in the vessel at culture temperature until harvest, and aggregate accumulates over that time. Galactosylation falls over the same period as the UDP-galactose that supplies the galactosyltransferase is depleted late in the culture. Afucosylation falls with duration as well, and the study has to bound that fall against the lower edge of the afucosylation criterion, below which the antibody loses effector function.

Culture temperature, Dissolved CO₂ (pCO₂) and osmolality are each scored against galactosylation and acidic charge variants, and temperature is scored against afucosylation as well. Galactosylation rises with culture temperature, which raises the specific activity of the galactosyltransferase, and falls as dissolved carbon dioxide and osmolality rise. Temperature is expected to act together with pH on the glycosylation attributes, since both act on the same glycosyltransferase activity in the culture, and an effect of that kind is estimated only by moving the two parameters together. The 5 parameters scored against a glycan, charge variant or aggregate attribute are therefore carried into one designed experiment.

The remaining 4 parameters change how many cells the culture reaches and how much antibody each of them makes, and none is scored against a quality attribute. Inoculation density, basal medium concentration and Nutrient feed-1 volume all raise the integral viable cell concentration over their ranges, and titer rises with it. Dissolved

oxygen raises titer through the productivity of the cells rather than through their number. Basal medium concentration is the one that needed argument. Platform experience shows a statistically significant but shallow effect on afucosylation, too small to move the attribute across its acceptance criterion over the concentrations the process uses, so the parameter is carried on process performance grounds and assessed one at a time.

Table 4.2: Production bioreactor: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Culture pH	pH excursion outside the characterized range (6.6-7.1)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)
Culture temperature	Temperature excursion outside 34-36 C	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)
Dissolved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Strong effect on galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Osmolality	Osmolality outside 360-440 mOsm	Affects galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA
Dissolved oxygen	DO outside 30-70%	Impacts peak VCD, titer and process consistency (no significant CQA impact)	process performance
Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	Drives peak/integral VCD and titer; no product-quality impact	process performance
Basal medium concentration	Basal medium concentration outside 0.8-1.6X	Statistically significant but shallow effect on afucosylation; drives titer	process performance

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	Slight effect on galactosylation; critical for titer	process performance

4.2 Harvest and Clarification

Harvest recovers and clarifies the culture and forms no product quality attribute. All 3 of its parameters are scored against process performance alone and all sit at the floor of the register at 96. No parameter of this step is carried into a designed experiment.

The failure modes are a centrifugation speed outside its range, a depth filter loaded beyond its capacity, and a high turbidity after clarification. A centrifugation speed below its range leaves more cells and cell debris in the centrate. A particle settles faster the stronger the centrifugal field acting on it, so a lower speed settles less of the fine debris in the time the material spends in the bowl, and what stays in the centrate then loads the depth filters. A depth filter loaded beyond its rated area passes particulate instead of retaining it, and the turbidity of the clarified harvest rises. Turbidity is measured on the clarified harvest, and particulate that reaches the capture column raises its back pressure, so a failure at this step is seen before the batch is released. The capture step is robust to the feedstock over the range this step produces.

Table 4.3: Harvest and clarification: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	Affects clarification/yield (no CQA impact)	process performance
Depth filter load	Depth filter load outside 80-160 L/m2	Affects filter capacity/turbidity (no CQA impact)	process performance
Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)	process performance

4.3 Protein A Chromatography

Capture binds the antibody from the clarified harvest and removes the bulk of the host cell protein and DNA. Protein load and elution buffer pH are the 2 parameters scored against a quality attribute, both against host cell protein at a severity of 6. A protein load above the characterized range raises the host cell protein in the pool, because more impurity is carried onto the column for the same volume of resin and the wash removes a smaller share of it. A lower elution pH raises it as well, since a more acidic elution buffer displaces host cell protein that the wash left bound to the resin. The two act together, and both are carried into the capture design.

The capture eluate is the load for the low pH hold. Elution buffer pH therefore fixes the pH of the pool before titration and the amount of acid needed to bring it to the hold set-point.

Load flow rate and end of pool collect are scored against step yield and against no quality attribute. Yield falls when a high load is run at a high flow rate, because the antibody has less time to diffuse into the resin pores and part of it passes through the column unbound. Yield rises as collection is carried further down the descending side of the elution peak, up to the point where the pool is diluted without gaining product. Both parameters are carried into the capture design, because the same design bounds yield and they cost no additional runs once the design exists. Operating temperature and bed height are assessed one at a time.

No operating parameter of this step is scored against leached Protein A, although the attribute is formed here. Leached Protein A enters the pool from the affinity ligand itself, and the amount released is set by the resin lot and by how many cycles the resin has run, both controlled under SOP-2008, rather than by any parameter in Table 5.2. The polishing steps clear what reaches them. The capture study measures leached Protein A in the pool across the designed experiment, so the ranges this assessment scopes are bounded against it even though no parameter is ranked on it.

Table 4.4: Protein A chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Protein load	Protein load above characterized range (>50 g/L resin)	Increases pool HCP (interaction with elution pH)	Host Cell Protein (HCP)
Elution buffer pH	Elution pH below characterized range (<3.2)	Increases pool HCP; also sets up viral inactivation feed	Host Cell Protein (HCP)
Load flow rate	Load flow rate high with high load	Reduces product yield (no CQA impact)	process performance
End of pool collect	End-of-pool collection outside 2.0-3.2 CV	Affects product yield (no CQA impact)	process performance

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Operating temperature	Parameter outside characterized range	No significant product-quality impact expected	process performance
Bed height	Parameter outside characterized range	No significant product-quality impact expected	process performance

4.4 Low-pH Viral Inactivation

Holding the pool at low pH inactivates enveloped virus by disrupting the lipid envelope. XMuLV needs to enter a cell, and the attribute register assigns the cumulative XMuLV clearance requirement to this step. Of its 4 parameters, 3 reach the top of the register at 700, which is the highest initial risk priority number in the assessment.

Inactivation of enveloped virus at low pH follows first order kinetics. Inactivation pH and temperature set the rate constant, and the hold time sets how long that rate acts, so the log reduction the step achieves is an extent and not a rate. A pH above the range therefore lowers the rate constant and leaves less virus inactivated over the same hold, and a hold shorter than its range leaves less inactivated at the same pH. A temperature below the range lowers the rate constant as well, and that matters most at the short end of the hold and the high end of the pH range, where the extent reached is smallest. The three parameters are assessed in one designed experiment because the extent depends on all three together and none of them can be bounded with the other two held at a single setting.

Inactivation pH carries a second failure mode in the opposite direction. A pH below the characterized range exposes the antibody to conditions under which aggregation and precipitation have been seen on the platform, and a hold longer than its range raises aggregate at any pH in the range.

Detection is at the top of the scale for each of the three. Neither an in-process control nor a release test on the drug substance measures the log reduction the hold achieves, and the claim is made instead from a spiking study on a qualified small scale model (International Council for Harmonisation 2023a).

A-Mab concentration is scored against process performance. Prior product experience shows no effect of concentration on the rate of inactivation, on aggregation or on charge variants over the range the pool reaches, so the parameter is assessed one at a time.

Table 4.5: Low-pH viral inactivation: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Loss of XMuLV inactivation assurance or increased aggregate	Viral clearance — XMuLV (enveloped), Aggregates (HMW)
Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Incomplete inactivation or increased aggregate	Aggregates (HMW), Viral clearance — XMuLV (enveloped)
Temperature	Temperature < 15 C	Lower inactivation rate at short time / higher pH	Viral clearance — XMuLV (enveloped)
A-Mab concentration	Concentration > 35 g/L	No significant effect on inactivation, aggregation or charge variants	process performance

4.5 Cation Exchange Chromatography

The cation exchange step is the principal aggregate reduction step of the process and also clears host cell protein, DNA and leached Protein A. It forms no quality attribute of its own. Of its 5 parameters, 4 are scored against a quality attribute, and protein load carries the highest severity of them at 8 because both aggregate and host cell protein are at risk.

Aggregate is the attribute that matters most here. No step after the cation exchange step is credited with aggregate reduction, so aggregate that leaves this step in the pool reaches the drug substance. The aggregate content of the pool rises with protein load, with elution buffer pH and with elution stop collect. Aggregate elutes after the monomer, so extending collection down the descending side of the peak pulls more of it into the pool, and a higher load leaves less resin per gram of antibody to separate the two. Load and elution pH act together, since the resolution between monomer and aggregate in the elution peak depends on both. Load and wash conductivity is scored against host cell protein, where a conductivity below the range weakens the wash and leaves more host cell protein bound to be eluted with the product.

Elution flow rate is scored against step yield. A higher elution flow rate broadens the elution peak, because the antibody has less time to diffuse out of the resin pores, and a broader peak dilutes the pool. Prior knowledge places no effect of flow rate on pool aggregate large enough to score, so the parameter is assessed one at a time.

Table 4.6: Cation exchange chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Protein load	Protein load above characterized range (>40 g/L resin)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	Aggregates (HMW), Host Cell Protein (HCP)
Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Reduced HCP clearance	Host Cell Protein (HCP)
Elution buffer pH	Elution pH outside 5.8-6.2	Alters aggregate distribution and yield	Aggregates (HMW)
Elution stop collect	Extended elution stop-collect (>1.5 OD)	Minor increase in pool aggregate	Aggregates (HMW)
Elution flow rate	Elution flow rate outside 100-300 cm/hr	Affects peak shape and yield (minor aggregate)	process performance

4.6 Anion Exchange Chromatography

The anion exchange step is operated in flow-through mode. At the load pH the step runs at, host cell protein, DNA and virus carry a net negative charge and bind to the resin, while the load pH sits below the isoelectric point of A-Mab, which therefore carries a net positive charge and passes through. The attribute register assigns the cumulative MVM clearance requirement to this step. All 5 of its parameters are scored against a quality attribute.

Load pH is scored against 3 attributes and reaches 700. It governs the charge on both the impurities and the resin, so a load pH below the characterized range weakens the binding of host cell protein and of both model viruses at once. Clearance also falls as the load conductivity rises, because the counter-ions in the load compete with virus and impurity for the charged groups on the resin. Load pH and load conductivity act together, since one sets the charge on the species and the other sets how strongly that charge is screened. Equil and wash-1 conductivity and protein load are scored against host cell protein, where the risk is a loss of binding capacity rather than a loss of clearance, since the mass of impurity presented to a fixed volume of resin rises with the load.

Operating flow rate is the one parameter in the register with a quality attribute at risk that is not carried into a designed experiment. A higher flow rate shortens the residence time of the load in the bed, which leaves a virus particle less opportunity to reach and bind a charged site on the resin. The parameter reaches the step through residence time alone, and prior knowledge places no interaction between residence

time and the load chemistry, so it is bounded by moving it one at a time. PCP-008 records that justification with the design.

Table 4.7: Anion exchange chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Load pH	Load pH below characterized range (<7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Equil/Wash 1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal	Host Cell Protein (HCP)
Load conductivity	Load conductivity above characterized range	Reduced viral clearance (interacts with pH)	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Protein load	Protein load above characterized range (>300 g/L resin)	Reduced HCP removal capacity (viral clearance robust)	Host Cell Protein (HCP)
Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)

4.7 Small-Virus Retentive Filtration

The virus filter removes small non-enveloped virus by size and is the principal MVM clearance step. Both of its parameters are scored against MVM clearance and both reach 700. The step carries no parameter scored against process performance alone.

Volumetric load is the parameter the step turns on. Log reduction falls as the volume filtered per unit area rises, because the retentive layer accumulates material and the probability of parvovirus passage rises with it. Filtration pressure is scored against both model viruses, and it sets the driving force across that accumulating layer. Prior knowledge places neither the size nor the direction of a pressure effect on log reduction, and the exposure is largest at high volumetric load, where the layer is thickest. Both parameters are therefore carried into the designed experiment rather than bounded one at a time.

Table 4.8: Small-virus retentive filtration: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Filtration volume (load)	Volumetric load above 105 L/m ²	Reduced MVM log-reduction (parvovirus breakthrough risk)	Viral clearance — MVM (parvovirus)
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)

4.8 Ultrafiltration / Diafiltration

The formulation step concentrates the antibody and exchanges it into the drug substance buffer. It forms and clears no quality attribute. All 3 of its parameters are scored against process performance and sit at the floor of the register at 96.

A diavolume count below the range leaves the buffer exchange incomplete, because each diavolume removes a fixed fraction of the buffer species left in the retentate and the residual falls with the number of diavolumes rather than with the time spent filtering. Flux rises with transmembrane pressure until the membrane reaches the pressure independent region, beyond which the polarized layer at the membrane surface thickens and flux stops rising, so a transmembrane pressure outside its range changes the processing time rather than the material. The final concentration is the drug substance concentration itself, which is measured directly and released against a specification, so a failure of this parameter is seen before the batch is released. Each is assessed one at a time.

Table 4.9: Ultrafiltration and diafiltration: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)	process performance
Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)	process performance
Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)	process performance

5 Characterization study assignment

The assignment is the output of this assessment. Of the 37 parameters, 22 are assigned to a multivariate design and 15 to univariate assessment. A designed experiment is run at 6 of the 8 unit operations. The other 2 are harvest and formulation, neither of which is scored against a quality attribute, and no designed experiment is to be run for them. Table 5.1 gives the split per unit operation with the plan and report that cover each.

Table 5.1: Characterization scope per unit operation, and the documents that cover it.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

The parameters assigned to a multivariate design are those judged able to act together with another parameter of the same unit operation. The judgement was made from platform and prior product experience and it is recorded per parameter in §4. The capture step carries 2 parameters into a design although they are scored against process performance alone, because the same design bounds step yield and their inclusion costs no additional runs.

The assessment marks 1 parameter univariate (justified), which is the substitution the A-Mab risk ranking and filtering table permits where the study can be justified (CMC Biotech Working Group 2009). It is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and reaches the step

through residence time alone. The justification is that a parameter which does not act together with the load chemistry can be bounded by moving it one at a time, and PCP-008 carries it with the design.

No parameter is excluded from study. The lowest ranking parameters change step yield and process consistency rather than a quality attribute, and they are studied one at a time so that the ranges entered in the batch record rest on data rather than on platform precedent alone.

Table 5.2 gives the parameter set handed to the plans, with the set-point, the range to be studied and the normal operating range. For 31 of the 37 parameters the range to be studied is wider than the normal operating range on both sides, so that the edges of the characterized region lie outside the region in which the process will run (CMC Biotech Working Group 2009). For the remaining 6 the two ranges share a bound, because that bound is a limit of the step rather than a control range. A minimum hold time and a maximum volumetric load are the clearest cases, and neither can be exceeded in the direction that would widen the study.

Table 5.2: The parameter set handed to the characterization plans, with the range to be studied.

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
3	Culture pH	pH	6.85	6.6-7.1	6.75-6.95	multivariate
3	Culture temperature	°C	35	34-36	34.5-35.5	multivariate
3	Dissolved CO2 (pCO2)	mmHg	100	40-160	70-130	multivariate
3	Osmolality	mOsm	400	360-440	375-425	multivariate
3	Culture duration	days	17	15-19	16-18	multivariate
3	Dissolved oxygen	% sat	50	30-70	40-60	univariate
3	Initial viable cell conc.	e6 cells/mL	1	0.5-1.5	0.8-1.2	univariate
3	Basal medium concentration	X	1.2	0.8-1.6	1-1.4	univariate
3	Nutrient feed-1 volume	% WV	12	9-15	10-14	univariate
4	Centrifugation (rcf)	g	9000	6000-12000	8000-10000	univariate
4	Depth filter load	L/m2	120	80-160	100-140	univariate
4	Post-clarification turbidity	NTU	5	1-20	1-10	univariate

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
5	Protein load	g/L resin	30	10-50	20-40	multivariate
5	Elution buffer pH	pH	3.55	3.2-3.9	3.4-3.7	multivariate
5	Load flow rate	cm/hr	200	100-300	150-250	multivariate
5	End of pool collect	CV	2.6	2-3.2	2.3-2.9	multivariate
5	Operating temperature	°C	22	15-30	18-26	univariate
5	Bed height	cm	20	10-30	15-25	univariate
6	Inactivation pH	pH	3.5	3.2-4	3.4-3.6	multivariate
6	Hold time	min	90	60-180	60-120	multivariate
6	Temperature	°C	21	15-25	19-23	multivariate
6	A-Mab concentration	g/L	20	5-35	5-35	univariate
7	Protein load	g/L resin	25	10-40	18-32	multivariate
7	Load/Wash conductivity	mS/cm	5	3-7	4-6	multivariate
7	Elution buffer pH	pH	6	5.8-6.2	5.85-6.15	multivariate
7	Elution stop collect	OD desc.	1	0.5-1.5	0.7-1.3	multivariate
7	Elution flow rate	cm/hr	200	100-300	150-250	univariate
8	Load pH	pH	7.5	7.2-7.8	7.35-7.65	multivariate
8	Equil/Wash-1 conductivity	mS/cm	2.6	1.6-3.6	2.1-3.1	multivariate
8	Load conductivity	mS/cm	5.5	3-8	4-7	multivariate
8	Protein load	g/L resin	200	50-300	120-280	multivariate
8	Operating flow rate	cm/hr	300	150-450	200-400	univariate
9	Filtration volume (load)	L/m2	90	50-140	50-105	multivariate
9	Filtration pressure	psi	13	8-30	10-20	multivariate

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
10	Number of diavolumes	DV	8	7-10	7-9	univariate
10	Transmembrane pressure	psi	12	10-15	10-14	univariate
10	Final DS concentration	g/L	75	65-85	70-80	univariate

The assignment is prospective. It is revisited when the characterization reports are available, and a parameter whose measured effect contradicts the judgement made here is reported as such in the report of its unit operation.

6 Assumptions and limitations

The assessment assumes that platform behaviour transfers to A-Mab. The mechanisms it relies on were established on the prior humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab and on A-Mab clinical and engineering experience, and the assumption is that A-Mab responds to the same parameters in the same directions (CMC Biotech Working Group 2009). Where it does not, the characterization study finds it, which is one of the reasons the studied ranges are wider than the operating ranges.

The parameter list is assumed complete. It is the list carried in the process description transferred under PTP-001, and a parameter that is not in that description is not assessed here. An equipment or facility difference at the receiving site that introduces a new parameter is handled under the transfer plan and would require this assessment to be revised.

Severity is taken from the quality attribute assessment and is not re-derived. That assessment scores impact and uncertainty and deliberately excludes manufacturing capability and detectability (CMC Biotech Working Group 2009), so a severity in Table 3.1 says what a failure would mean for the patient and says nothing about how likely the process is to produce one.

Occurrence and detection are assigned by class rather than parameter by parameter, so the initial risk priority number takes only 5 values across 37 parameters. The number therefore separates classes of parameter well and separates parameters within a class not at all. Ranking within a unit operation should be read from severity and from the reasoning in §4.

The interaction judgement is the weakest part of the assessment. A designed experiment estimates interactions only among the parameters it carries, so an interaction between a parameter in the design and one left out of it cannot be found by that design. The clearest exposure is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and studied one at a time. If the anion exchange study finds that residence time acts together with the load chemistry, the univariate result for flow rate would not bound the step and the design would have to be augmented.

Interactions across unit operations are not scored. Each parameter is assessed against the attributes of its own step. Material does carry properties forward, and the elution pH of the capture step setting the starting pH of the low pH hold is the clearest case, but such links are handled by the in-process limits of the receiving step rather than by a score in Table 4.1.

Finally, no number in this document is a measurement of A-Mab process behaviour. The scores are judgements, the studies are not yet run, and the ranges in Table 5.2 are the ranges to be studied and not ranges that have been shown to be acceptable.

7 Outputs and downstream use

This assessment produces three things. The ranking in Table 4.1 is the pre-characterization risk basis. The study type in Table 4.1 and the summary in Table 5.1 are the scope. The parameter set in Table 5.2 is what each plan instantiates.

Per unit operation the consumer is the characterization plan named in Table 5.1. Each plan cites this assessment as its risk basis, takes the study type and the range to be studied, and adds the scale-down model, the analytical methods, the sampling plan and the acceptance criteria. The report of that unit operation then measures the effects, defines the operating region and classifies each parameter under SOP-4001.

PCMP-001 consumes the assessment at campaign level. It states once what the individual plans would otherwise each restate, including the count of parameters going to multivariate and to univariate study across the train. PCMR-001 consolidates the reports and is where the risk that remains after characterization is addressed.

This document is superseded by its own revision. A change to the parameter list, to a severity, or to an assigned study type is made by revising this assessment, and the affected plans are revised with it.

8 Approvals

This assessment is approved by the functions listed in Table 1. Approval confirms that the parameter list is complete against the transferred process description, that the severity carried for each attribute agrees with the quality attribute assessment, and that the study type assigned to each parameter is supported by the reasoning given in §4. A change to the parameter list or to an assigned study type after approval is made by revision of this document.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

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